ORGANIC FARMING OR SUSTAINABLE AGRICULTURE

LAST 5 YEAR PREVIOUS YEAR QUESTION (2019-2023)

1. Which one of the following is the coordinating agency for organic production and exports of organic products in India? (2021)

a. APEDA
b. NHB
c. Department of industry and trade
d. All of the above

2. Under which farming system the use of natural pest controls and biofertilizer is promoted and the use of any type chemical and synthetic pesticides, fertilizer is completely avoided? (2022)

a. Precision farming
b. Companion cropping
c. Organic farming
d. All of the above

3. Which one of the following is first fully certified organic state in India? (2021 mains)

a. Nagaland
b. Sikkim
c. Tamil Nadu
d. Madhya Pradesh.

4. Among which state _________has covered largest area under organic farming. (2021 mains)

a. Uttar Pradesh
b. Madhya Pradesh
c. Sikkim
d. Maharashtra

5. Which of the following scheme has its main objective to promote organic farming in India and production of Agriculture products free from Chemicals and Pesticides residue by adopting ecofriendly with low-cost technologies? (2021 mains)

a. PKVY-parampragat Krishi vikas yojana
b. PM krishi sinchayee yojana
c. PM Fasal bima yojana
d. All of the above

6. Which of the following are the main components of the organic farming? (2021 mains)

a. Use of no chemical fertilizer, pesticide, weedicide etc.
b. Under organic farming the system will reach to stability after few years.
c. Promotion of use of green manure, locally available seed, natural pesticide as control to pest.
d. All of the above

ANSWERS

1. A 2.C 3.B 4.B 5.A 6.D

ORGANIC FARMING OR SUSTAINABLE AGRICULTURE

Sustainable agriculture is a **balanced management system of the renewable resources including soil, wildlife, forest, crops, fish, livestock, plant genetic resources and ecosystem without degradation** and to provide food, livelihood for the current and the future generations maintaining or improving productivity and ecosystem services.

- ✓ Sustainable agriculture is also known **as Eco farming or organic farming** or natural farming or permaculture.
- ✓ Sustainable agriculture is **considered as integrated, low input and highly productive farming system**.

Need Of Sustainable Agriculture/Organic Farming

There are various problems **in modern agriculture which is not sustainable for a longer period of time – the problem in modern agriculture** is given below

PROBLEMS IN MODERN AGRICULTURE

1. Faulty agriculture practices lead to the adverse effects such as **loss of fertile top soil** through wind and water erosion.
2. Due to improper management of irrigation water**, salinization and alkalization** of the soil takes place.
3. Approx. 6 m ha of land is affected by **water logging** and another 7 m ha is **salinized** due to faulty irrigation practices.
4. Monocropping or especially **rice wheat** cropping system led to the loss of fertility of the soil and impact the biodiversity of that region.
5. Excessive use of chemical fertilizer, pesticide, weedicide causes **pollution of water bodies and ground water resources**.
6. Excessive use of ground water for irrigation with faulty irrigation practices causes **depletion of the ground water table**.
7. Development **of pest resistance and weed resistance** due to excessive use of chemical pesticide and weedicide.
8. **Loss of biological diversity** and erosion of germplasm resources through removal of natural flora and fauna.

9. Land degradation due to excessive shifting cultivation in north eastern region.

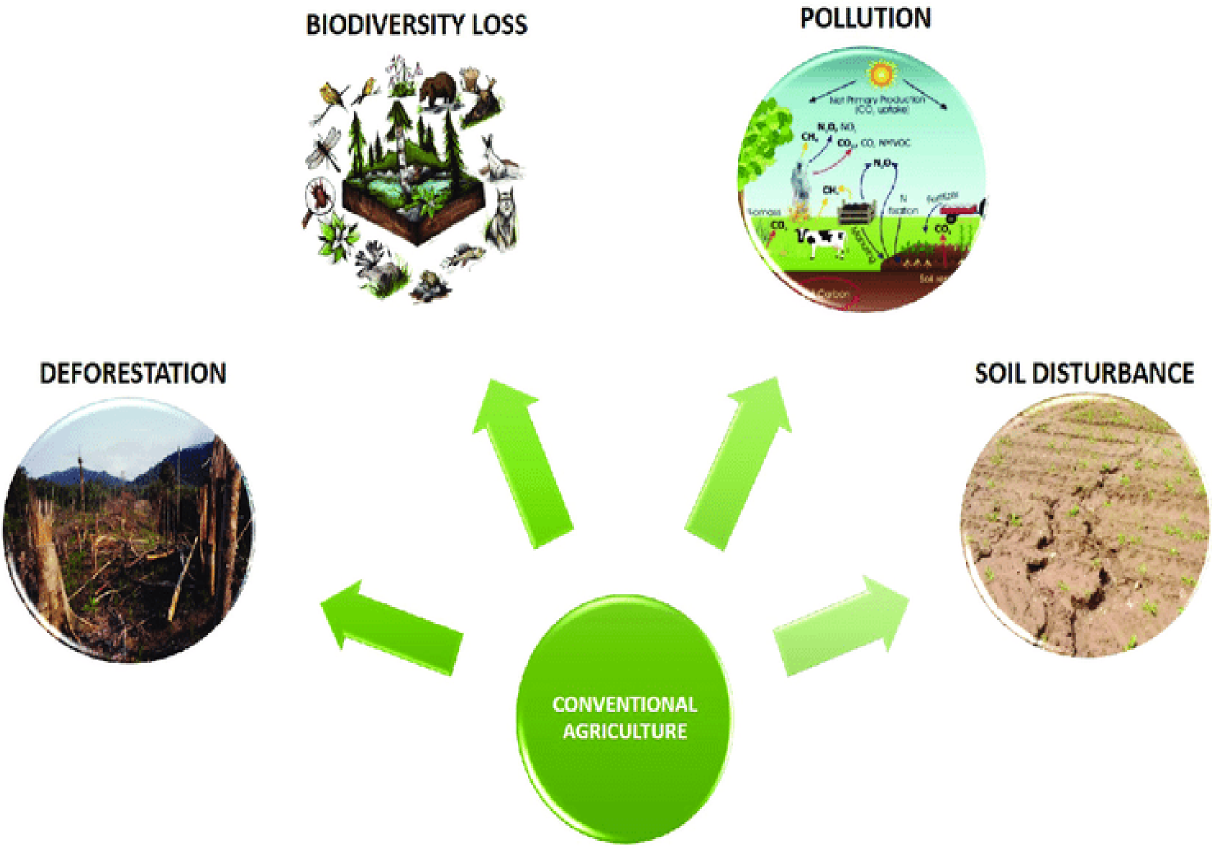


Image (I)- Problem In Conventional/Modern Agriculture System

DIFFERENCE BETWEEN SUSTAINABLE AGRICULTURE AND MODERN AGRICULTURE

The main difference between the two is that-

Modern Agriculture-

- ✓ Use synthetic **fertilizer NPK** to enrich the soil.
- ✓ **Chemical pesticide and weedicide** used to control weed and pest

Sustainable Agriculture-

- ✓ Involves agroforestry.
- ✓ Multilevel cultivation.
- ✓ Integrated animal husbandry

Table 1 difference between sustainable agriculture and modern agriculture

Particulars	**Sustainable Agriculture**	**Modern Agriculture**
Plant Nutrients	**Farmyard manure, compost, green manure** and crop rotation is used	**Chemical fertilizers like NPK**
Pest Control	**Cultural method, crop** rotation and biological method	Toxic chemical is used
Inputs	**High diversity, renewable** and biodegradable inputs are used	**High productivity and low diversity** chemicals are used
Ecology	Stable ecology	Fragile ecology
Use Of Resources	The **rate of extraction of resources does not** exceeds rate of regeneration	**Rate of extraction exceeds rate of regeneratio**n like overgrazing, deforestation etc.
Quality Of Food	**Longer shelf like** and does not contain harmful toxic residues	**Less shelf life** and contains various toxic chemical compounds.

MANAGEMENT PRACTICES IN ORGANIC FARMING

The management practices are mainly aimed at **obtaining sustainable production with limited or no chemical inputs** with preference to farm grown inputs without pollution and minimum damage to natural resources.

These are the following important steps in sustainable agriculture

1. **Water Shed Management**- for dryland agriculture conservation of soil and moisture conservation measures, land use based on land capability, afforestation and efficient crop production.

Image (ii)-Various Components of Organic Farming

2. **Conservation Of Genetic Resource**-land races are to be preserved for future generations.

3. **Tillage**- minimum tillage practices and **maintenance of top 8 cm of soil** is vital because most of the biological activities, microorganism and organic matter is present in this layer.

4. **Nutrient Management**- use of green manure like sun hemp, **dhaincha, are grown and incorporated in soil after 40 days, use of biofertilizer like azolla, blue green algae,** azotobacter, rhizobium, etc. crop rotation should be practiced with introduction of a legume crop.
5. **Efficient Water Management**- it **can be subdivided** into rain water management and irrigation water management.

- ✓ **Rain water management**- water harvesting, supplemental irrigation, reducing evapotranspiration etc.
- ✓ **Irrigation water management**- scheduling irrigation at appropriate time with adequate quantity of water.

6. **Weed Management**-can be controlled by using **cultural, physical, biological methods, crop rotation will break crop weed cycle, tillage** and hand weeding.

7. **Pest Management**- integrated pest management is to be used **and plant-based material like neem oil, microbial agents, natural enemy** of that pest is to be used as a pest control.

8. **Crop Rotation**-it is very important aspect of organic **farming as it will improve soil fertility, weed, insect and disease management. Legumes ar**e very essential in crop rotation and legumes are to be grown **on 30 to 50% of the crop land.**

ADVANTAGES OF SUSTAINABLE AGRICULTURE

- ✓ Maintenance of **ecological balance.**
- ✓ **Low-cost cultivation**.
- ✓ **Clean environment and nutritious** food without pesticide residue.
- ✓ **Improve fertility of the soil**.
- ✓ Quality of **the produce will have longer shelf** life.

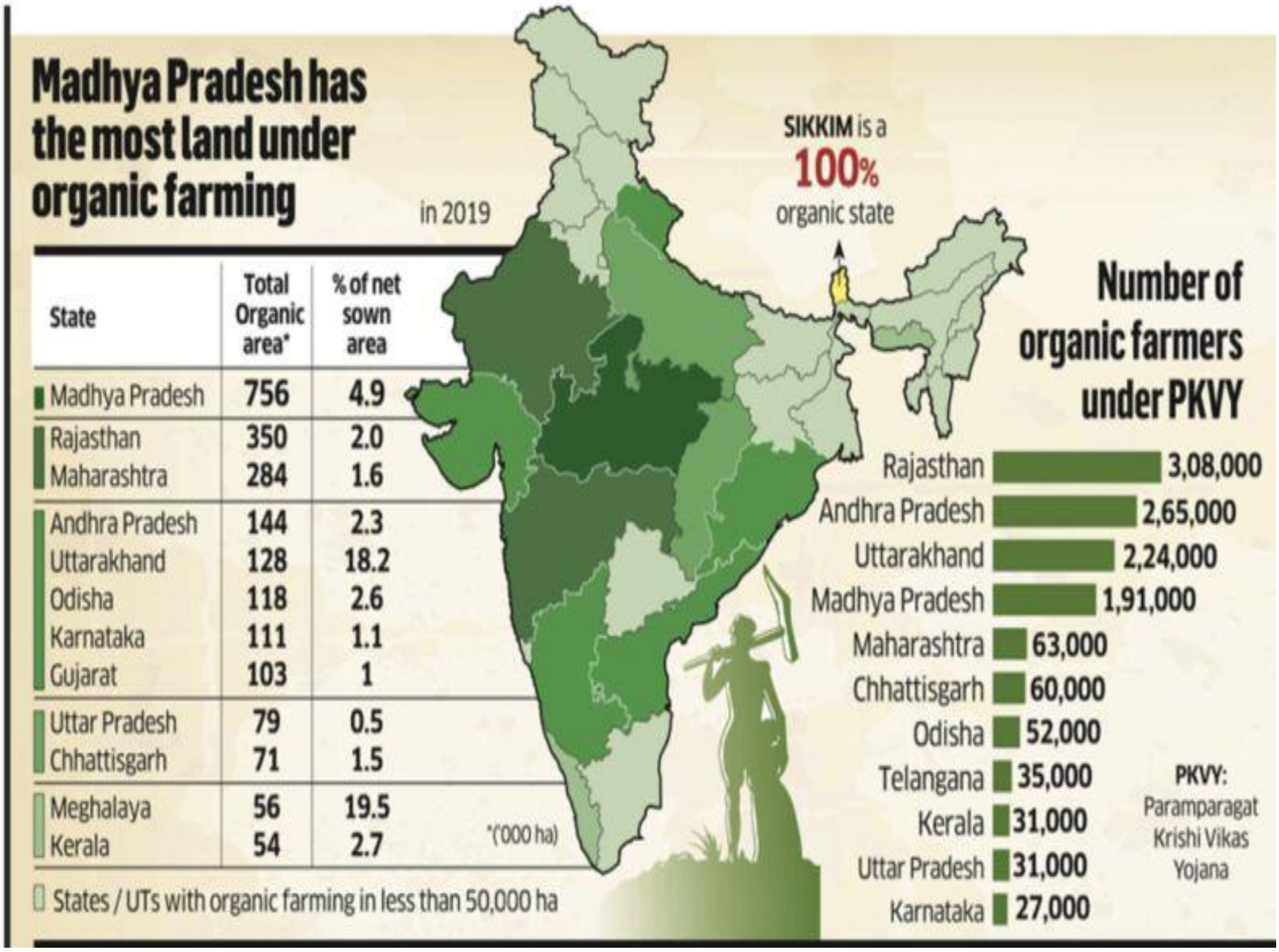
DISADVANTAGE OF SUSTAINABLE AGRICULTURE

- ✓ Low yield as compared to modern agriculture.
- ✓ Lack of timely and effective control of weeds, pest and disease.

✓ The **conversion process from modern agriculture to sustainable agriculture** will take three to six years to reach equilibrium.

ORGANIC FARMING IN INDIA

India has the highest number of organic farmers in the **world at 44.3 lakhs, and 59.1 lakh ha area** has been brought under organic farming by 2021-22, says the Survey. Organic and natural farming provides chemical and pesticide free food grains and crops, improves soil health and reduces environmentally pollution.



The Government has been promoting organic farming by three dedicated schemes viz.,

1. Paramparagat Krishi Vikas Yojana (PKVY) - Under PKVY, 32,384 clusters **totaling 6.4 lakh ha area and 16.1 lakh farmers** have been covered as of November 2022

2. **Mission Organic Value Chain Development for North Eastern Region (MOVCDNER)** through cluster and Farmer Producer Organisations formation-under **MOVCDNER, 177 FPOs/FPCs have been created, covering 1.5 lakh farmers and 1.7 lakh hectares** to promote organic farming of niche crops in the North East Region.

3. **Under Bhartiya Prakratik Krishi Paddhati (BPKP)**, a scheme to help farmers adopt all forms of traditional/ecological farming practices including Zero-Budget Natural Farming (ZBNF), 4.09 lakh hectares of land have been brought under natural farming in eight states.

Sikkim was the first state to be certified as fully organic state.

Madhya Pradesh has the largest area approx. 68 million hectares under organic farming followed by Maharashtra, Gujarat, and Rajasthan.

ZERO BUDGET NATURAL FARMING

Natural Farming offers a solution to various problems, **such as food insecurity, farmers' distress, and health problems arising due to pesticide and fertilizer residue in food and water, global warming, climate change and natural calamities**. It also has the potential to generate employment, thereby stemming the migration of rural youth. Natural Farming, as the name suggests, is the art, practice and, increasingly, the **science of working with nature to achieve much more with less.**

COMPONENTS OF ZERO BUDGET NATURAL FARMING

1. Beejamrit

It is used for seeds, seedlings or any planting material. It is effective in protecting young roots from fungus. **Beejamrit is a fermented microbial solution, with loads of plant-beneficial microbes, and is applied as seed treatment**. It is expected that the beneficial microbes would colonize the roots and leaves of the germinating seeds and help in the healthy growth of the plants.

Inputs needed: 5 kg cow dung, 5 litre cow urine, 50-gram lime, 1kg bund soil, 20 litre water (for 100 kg seed)

Application as a seed treatment: Add Beejamrit to the seeds of any crop; coat them, mixing by hand; dry them well and use them for sowing. For leguminous seeds, which may have thin seed coats, just dip them quickly and let them dry.

2. Jivamrit-

Jivamrit acts as a **bio stimulant** by **promoting the activity of microorganisms in the soil and also the activity** of **phyllo-spheric microorganisms** when spayed on foliage. It acts like a primer for microbial activity, and also increases the population of native earthworms.

Inputs needed: 10 kg of fresh cow dung, 5-10 litre cow urine, 50-gram lime, 2 kg jaggery, 2 kg pulses' flour 1 kg uncontaminated soil and 200 litres water

Application of Jivamrit: This mixture should be applied every fortnight. **It should be either sprayed directly on the crops or mixed with irrigation water.** In the case of fruit plants, it should be applied on individual plants. The mixture can be stored for up to 15 days.

3. Mulching

Is defined as **covering of soil surface using both live crops and straw (dead plant biomass**) to conserve moisture, lower soil temperature around plant roots, prevent soil erosion, reduce runoff and reduce weed growth.

There are two types of mulches:

- ✓ **Crop Residue Mulch:** This comprises any dried vegetation, farm stubble, such as dried biomass waste etc. It is used to cover the **soil against severe sunlight, cold, rain etc. Residue mulching also saves seeds from birds, insects**, and animals.

- ✓ **Live Mulch:** Live mulching is practised by **developing multi-cropping/inter cropping patterns of short durational crops in the rows** of a main crop.

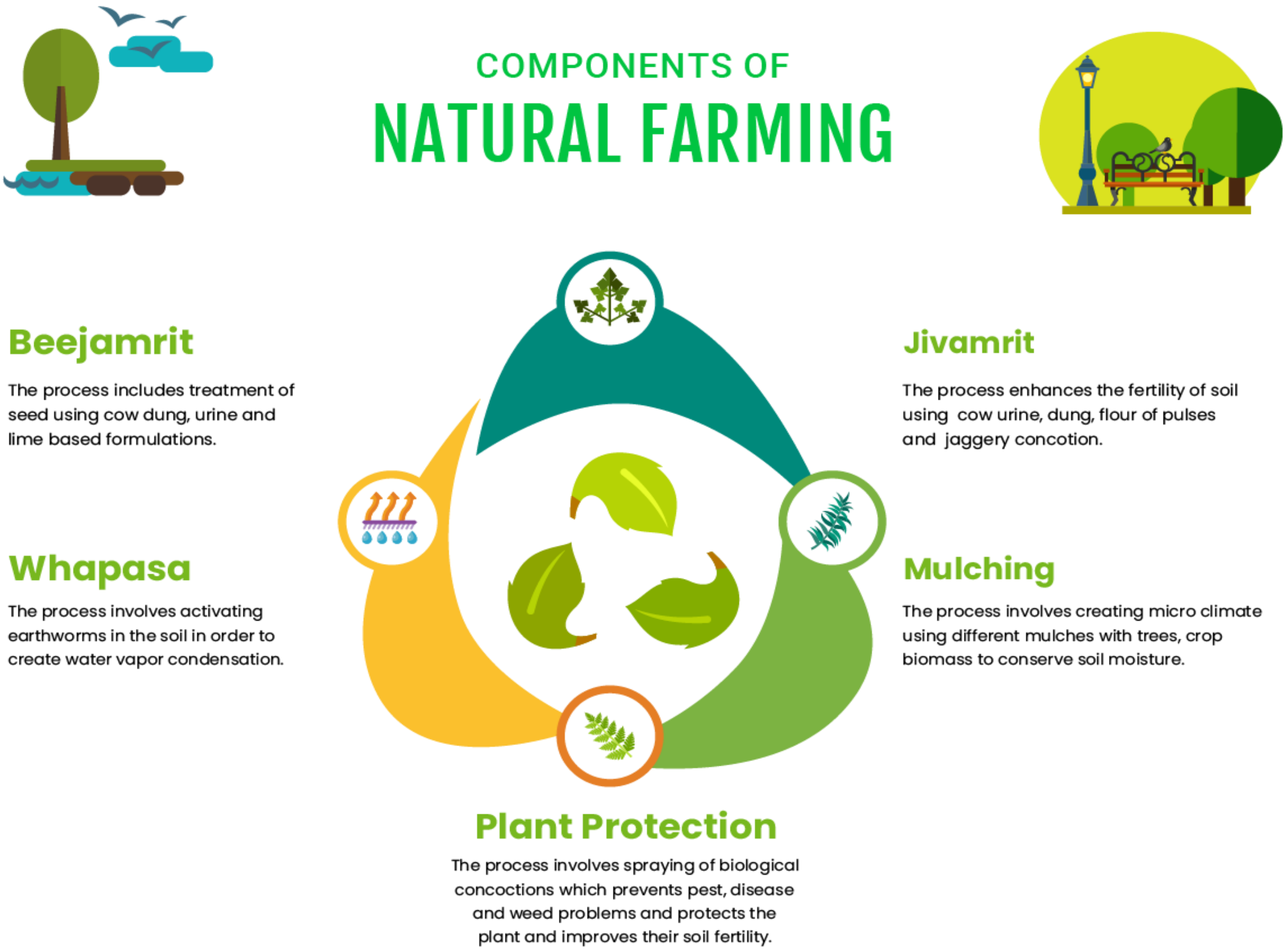


Image (iii)- components of zero budget natural farming

4. Whapasa

Means the **mixture of 50% air and 50% water vapour** in the cavity between two soil particles. It is the soil's microclimate on which soil organisms and roots depend for most of their moisture and some of their nutrients. It increases water availability, enhances water-use efficiency and builds resilience against drought.

5. Plant protection-

This is a **natural insecticide prepared from leaves which have specific alkaloids to repel pests**. It controls all sucking pests and hidden caterpillars that are present in pods and fruits.

- ✓ **Fungicide** prepared with cow milk and curd is found to be very effective in controlling the fungus.
- ✓ **Neemastra** is used to prevent or cure diseases, and kill insects or larvae that eat plant foliage and suck plant sap.
- ✓

Natural farming is practiced in various states in India

There are several states practicing Natural Farming. **Prominent among them are Andhra Pradesh, Chhattisgarh, Kerala, Gujarat, Himachal Pradesh, Jharkhand, Odisha**, Madhya Pradesh, Rajasthan, Uttar Pradesh and Tamil Nadu.

Andhra Pradesh has the largest area under natural farming followed by Gujarat and Madhya Pradesh

Map 2- States Practicing Natural Farming

Advantages Of Natural Farming

1. Farmers practising **Natural Farming reported similar yields** to those following conventional farming.
2. As **Natural Farming does not use any synthetic chemicals**, health risks and hazards are eliminated. The food has higher nutrition density and therefore offers better health benefits.
3. **Natural Farming aims to make farming viable and aspirational by increasing** net incomes of farmers on account of cost reduction, reduced risks, similar yields, incomes from intercropping.
4. Natural Farming aims **to drastically cut down production costs.**
5. **Natural farming generates employment on account of natural farming** input enterprises, value addition, marketing in local areas, etc.
6. Eliminate the **use of synthetic fertilizers which degrade the soil quality**.
7. Natural Farming ensures **better soil biology, improved agrobiodiversity and a more judicious usage of water with much smaller carbon** and nitrogen footprints.
8. By working with diverse crops **that help each other and cover the soil to prevent unnecessary water loss through evaporation, Natural Farming** optimizes the amount of 'crop per drop'.
9. The most **immediate impact of Natural Farming is on the biology of soil**—on microbes and other living organisms such as earthworms.
10. The **integration of livestock in the farming system plays an important** role in Natural farming and helps **in restoring the ecosystem**.

ORAGNISATION RELATED TO ORGANIC PRODUCE AND PRODUCTS

Organic Product/Produce

Any form of food or raw material (grain, species, fruit, oil etc) **produced or processed in a manner that are close to natural process** and free from any type of chemical residue is called as organic food/product/produce.

Organisation

APEDA- The Agricultural and Processed Food Products Export Development Authority (APEDA) was established by the Government of India under the

Agricultural and Processed Food Products Export Development Authority Act passed by the Parliament in December, 1985. **The Act (2 of 1986) came into effect from 13th February, 1986**

APEDA Comes under ministry of commerce and industry.

Head office: New Delhi

The Main Functions of APEDA Related to Organic Farming

- ✓ APEDA is the main agency **related to the production, export and certification** of the organic products in India.
- ✓ Promoting the **export-oriented production** and development of organic agri-products.
- ✓ **Implementing agency of national programme for organic production** and participatory guarantee scheme- which set standards for the organic products and certifies them as an organic product.

Certification of the organic products

Government of India launched an initiative of **national standard for organic production for which aimed at regulating production/ processing** and facilitating sale of quality and original organic products.

1. **National Standards for Organic Production: National Standards for Organic Production (NSOP)—** This body sets out the standards to be followed in the **cultivation, harvest, production, processing and trading of organic products.**

- ✓ The Standards for Organic Production are **notified in National Programme for Organic Production (NPOP)** by the Director General of Foreign Trade under the Foreign Trade (Development and Regulation) Act, 1992.

- ✓ The NPOP has a system where **in Certification Bodies are accredited with National Accreditation Body (NAB)** which is chaired by the Additional Secretary in the Department of Commerce.

✓ **The Secretariat of the NAB is with APEDA** - NPOP covers standards for crops and their products, live stocks and poultry products, aquaculture, apiculture etc. The exports from the country are as per the provisions in NPOP.

Certification Agencies: There are many agencies designated as certification agencies in the country. Some are **government owned and others are franchise of international certification agencies**. All the certifications whether they are done by a government or by a non-government organisation are accredited as per the NPOP and APEDA recommendations.

Certification System

India has two certification system for the organic products both the system is based on common standards but adopt different verification process.

1. **National programme for organic products- for export related standards for organic products.**

2. **Participatory Guarantee System (PGS)for India: for domestic and local market.**

National Programme For Organic Products-it is a kind of third-party **certification in which the farm and the processing of the agriculture produce is certified in accordance with national and international standards NPOP certification is facilitated by APEDA** - The Agricultural and Processed Food Products Export Development Authority under ministry of commerce and industry.

➢ **For export oriented Agri products**

Jaivik Bharat

Image (iv)-Jaivik Bharat logo for NPOP certified products

<u>Participatory Guarantee System For India (PGS)-</u>PGS India is locally focused quality assurance system which is built of foundation of trust, social network and knowledge exchange. Under this system **people in similar situation access, inspect and verify the production practice of each other and collectively declare entire holding as organic**. PGS for India is **facilitated by ministry of agriculture and farmers welfare through national centre for organic farming as its secretariat.**

Image (v)-logo for organic products certified under PGS Inida

A total of 6.12 lakh hectare land and 9.3 lakh farmers are covered under PGS India certification and around 15 lakh farmers covered under NPOP.

<u>National Centre For Organic Farming (NCOF):</u> This is a training institute under the Ministry of Agriculture and Farmers Welfare, Government of India.

<u>There are six regional centres</u> at Bengaluru, Bhubaneswar, Hisar, Imphal, Jabalpur and Nagpur.

<u>The Main Functions of NCOF Are</u>

- ✓ **<u>Promotion of organic farming in the country through technical capacity building of all stakeholders</u>** including human resource development, transfer of technology, promotion and production of quality organic and biological inputs.
- ✓ **<u>Certification of organic products</u>**. NCOF is also involved in awareness generation, capacity building, training and demonstrations, providing financial support to organic inputs and market development for farmers.